

Particulate Matter Monitoring & Health Effects

Issue 2026-09-18 | Entry window 18 September 2026 | 17 records screened in scope

17RECORDS IN SCOPE
(43 EXCLUDED)**8**OF 10 SUBTOPIC
BANDS POPULATED**6**EFFECT ESTIMATES
(ONLY 2 WITH A CI)**3**TIER-A
RECORDS**4**RECORDS CARRIED ON
METADATA ALONE

Signal of the day

Three independent records today attack the same assumption from three directions: that a surface-sited, mass-reporting monitor measures the exposure that matters. Source, height and composition each break it.

Source. Zhang et al. (*Nat Commun*) separated wildfire-specific from non-wildfire PM_{2.5} across **20 US states, 2006–19**, in a self-controlled design. Per 1 µg/m³, wildfire PM_{2.5} raised cardiopulmonary hospitalisation risk **10% (heart failure) to 16% (asthma)**; non-wildfire PM_{2.5} raised it **4.7% (COPD) to 8.5% (hypertension)**. Equal mass, unequal hazard, by roughly a factor of two. **Height.** Bai et al. (*Atmos Meas Tech*) used Doppler wind lidar over Hefei to show two dust episodes with opposite surface signatures: in one, a descending low-level jet drove aerosol down and surface PM₁₀ peaked at **410 µg/m³**; in the other, a stable jet core held the layer aloft and the surface peak arrived ~8 h after frontal onset. A surface network reads clean air while a loaded layer sits overhead. **Composition.** Yu et al. (*Atmos Chem Phys*) found marine organic-aerosol light absorption over the Yellow and Bohai Seas rising with both organic-nitrogen concentration and ON/OC ratio — absorption capability per unit carbon changes with continental outflow, which is precisely the term a fixed optical calibration assumes constant. None of the three is a sensor paper. Together they are the strongest argument this watch has carried for why the next increment in exposure assessment is source- and height-resolved, not denser mass.

What changed today

- ▶ **A large indoor-monitoring design paper, with a deployment protocol worth stealing.** Kim et al. (*Environ Health*) report the design and early findings of a residential indoor PM_{2.5} campaign in **130 older adults with idiopathic pulmonary fibrosis** in Seoul, 2021–23: MicroPEM runs of five days in each of four seasons, matched outdoor sampling, 36 chemical components, and infiltration factors of **51–70%** from sulfur, nickel and iron indoor/outdoor ratios. The operational contribution is non-contact redeployment by postal mail — the unglamorous problem that decides whether an unattended network survives a year.
- ▶ **The issue's evidence base is unusually thin on quantitative precision.** Six effect estimates were extractable and **only two carry a confidence interval**. Both of those are hazard ratios below the null from one UK Biobank analysis; the four estimates above the null — all from the wildfire paper — are point values with no interval reported in the abstract. Read the forest plot as direction only.
- ▶ **Four records are carried on metadata alone, and for the first time the gap is not purely Elsevier's.** A BMC deposit (*Part Fibre Toxicol*, PM and human embryonic stem cell osteogenic differentiation) and an ACS deposit (*Environ Sci Technol*) joined the usual two Elsevier same-day deposits with no abstract recoverable from Crossref, Europe PMC or PubMed. No finding is asserted for any of the four.
- ▶ **A useful null, and a reason to distrust it.** Temporao et al. found *no* robust day-to-night association for PM_{2.5} or NO₂ with EEG-measured sleep in a Budapest panel, while day-1 ozone predicted lower day-2 sleep efficiency. But exposure is city-aggregated from fixed stations: within-city contrast is near zero and misclassification is far worse for spatially heterogeneous PM_{2.5} than for ozone. The pollutant that survived is the one the exposure model measures best.
- ▶ **Two *Environ Sci Technol* Perspectives in one issue argue air into the exposome programme.** Sunyer-Caldú proposes a multi-matrix framework with a unified, toxicokinetically normalised exposure scale; Godri Pollitt's companion piece is titled "Air as an Integrative Matrix". Two of these on the same day is an editorial position, not a coincidence — though the first has no data and the second has no recoverable abstract.
- ▶ **Consensus returned nothing admissible for the second consecutive issue.** Ten sensor-calibration papers came back; five were already in the archive on a title match and all five survivors failed a Crossref created-date check, dating from January to June 2026. Scholar Gateway, queried for an abstract, again returned only off-topic Wiley content.

Corpus analytics

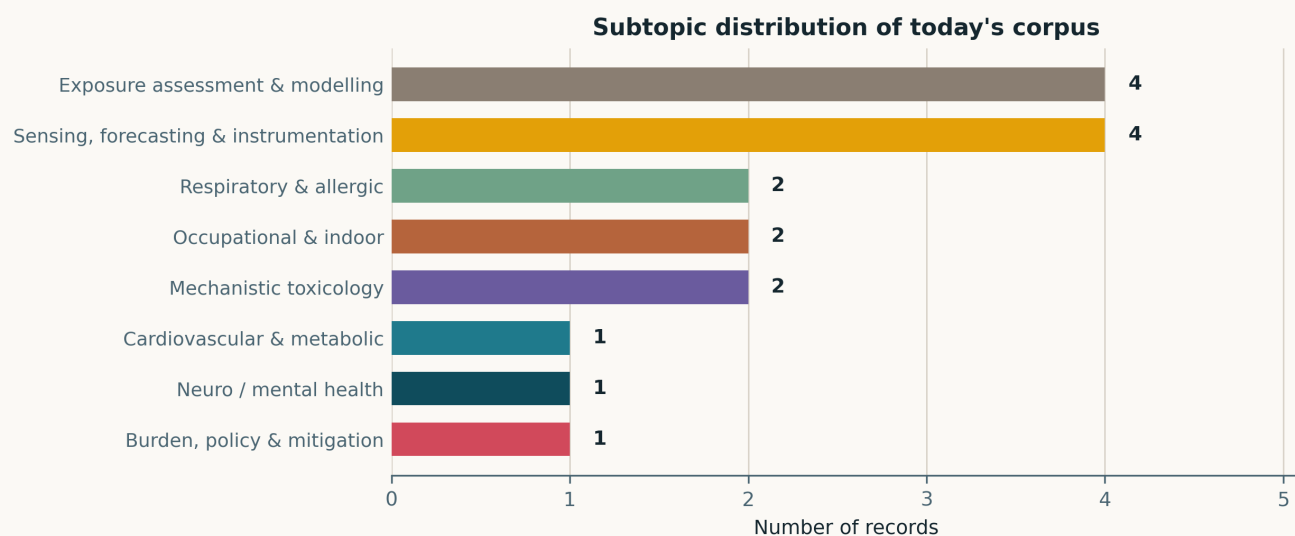


Fig. Subtopic assignment of the 17 in-scope records; eight of the ten bands are populated. The distribution is unusually measurement-weighted for this watch: *Exposure assessment & modelling* and *Sensing, forecasting & instrumentation* tie at four records each and together hold 8 of 17, while three bands hold two records and three hold one. *Reproductive & developmental* and *Other clinical endpoints* are empty today.

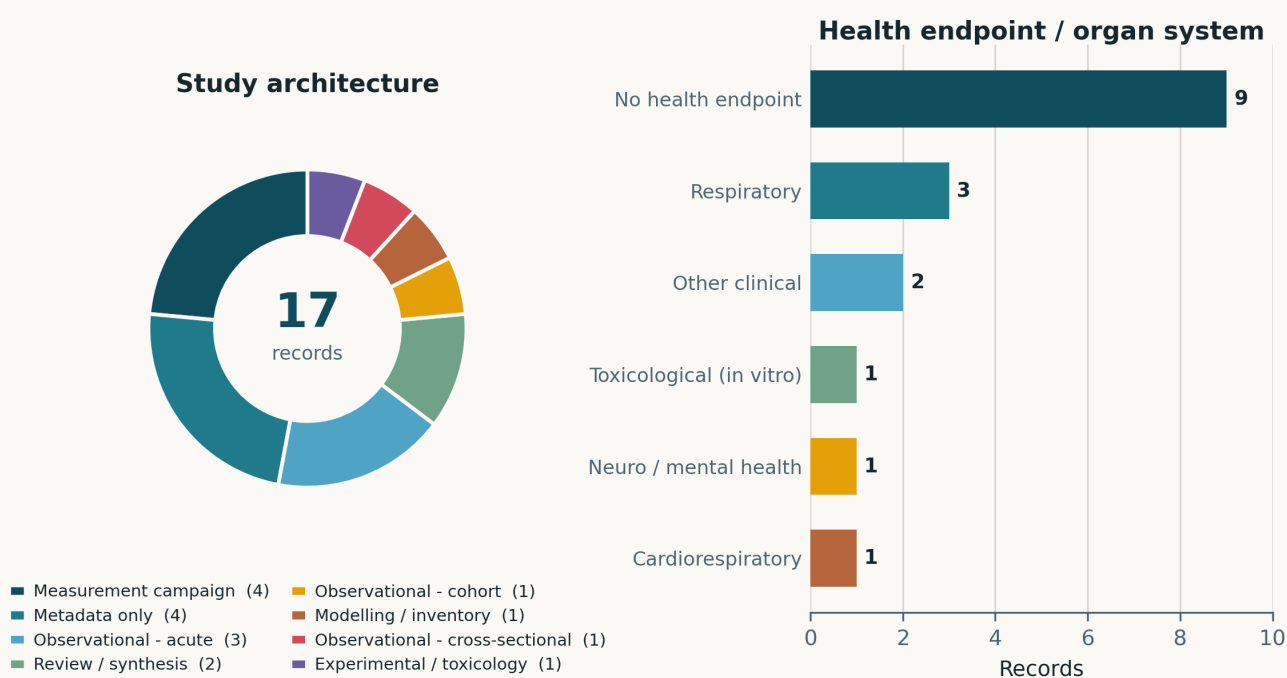


Fig. Left — study architecture. Measurement campaigns and metadata-only records tie as the largest groups at four apiece, which is the shape of a harvest dominated by same-day journal deposits. Primary observational epidemiology contributes five (three acute, one cohort, one cross-sectional) and evidence synthesis two. Right — health endpoint. “No health endpoint” is by far the largest bucket at nine of seventeen — the aerosol-chemistry, emissions, source-policy and not-applicable records all canonicalise into it — so more than half the issue reports no human outcome at all. Respiratory leads the clinical endpoints with three.

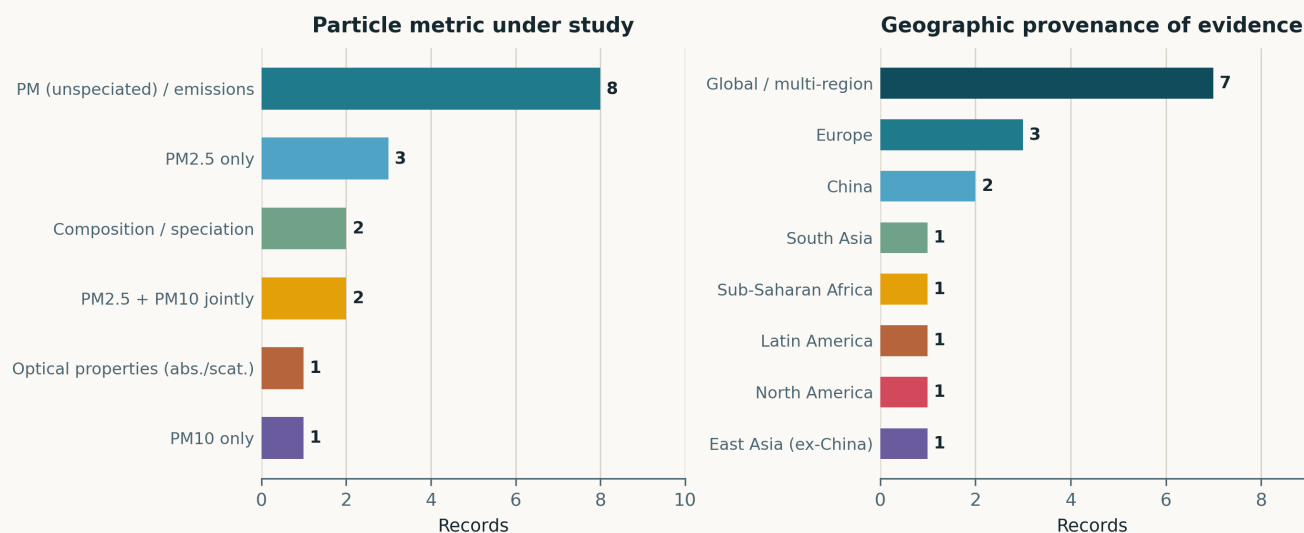


Fig. Left — particle metric. Unspecified PM or emissions terms cover eight records, and the breakdown is the point: **three are carried on metadata alone, three measured no particulate matter at all** (the greenness cohort, the bioinformatic analysis and the exposome Perspective), and two report a real but unharmonised PM term — pooled cement dust across eighteen studies, and industrial stack PM. Speciated composition appears twice and optical properties once. Right — geographic provenance. The *Global / multi-region* bar is seven, but that number is not what it looks like: only three are genuinely multi-region, and **four are records whose study location is not recoverable** — two same-day deposits with no abstract, one bioinformatic analysis with no field site, and one industrial modelling study that never names its plant. Europe contributes three, China two.

Life-course windows addressed today (records may span several stages)

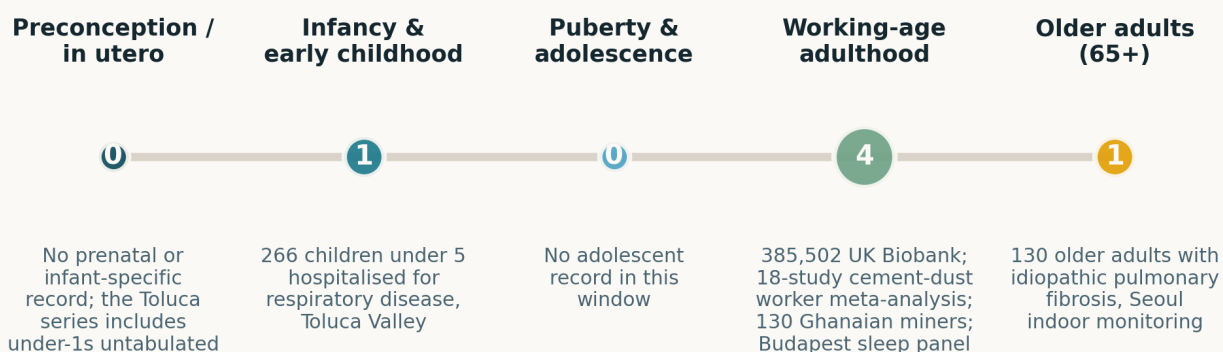


Fig. Life-course windows addressed; counts are non-exclusive. Working age carries four records — the UK Biobank COPD cohort, the 18-study cement-dust worker meta-analysis, the Ghanaian mining communities and the Budapest sleep panel. The paediatric entry is the Toluca series of 266 children under five; because bronchiolitis cases in that series are necessarily infants but no infant stratum is tabulated, the prenatal/infant window is recorded as zero rather than inferred.

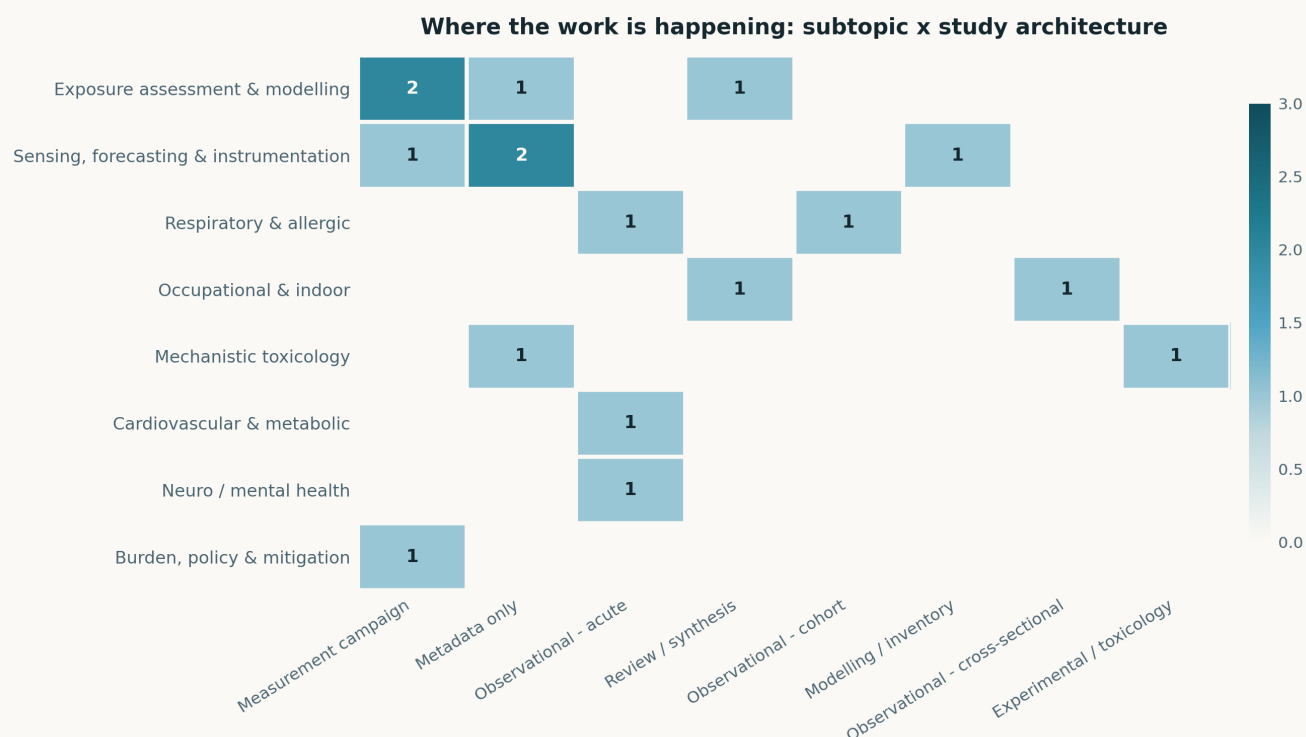


Fig. Subtopic × study-architecture cross-tabulation. Two structural features are worth naming. First, the *Exposure assessment & modelling* row carries **no health endpoint whatsoever** across its four records — two measurement campaigns, one Perspective and one metadata-only entry — and the *Sensing* row likewise; instrumentation and epidemiology are again unlinked within a single day's harvest. Second, *Mechanistic toxicology* is **not** wholly experimental this issue: one of its two records is a bioinformatic analysis in which no particle was ever administered, and the other is carried on metadata alone.

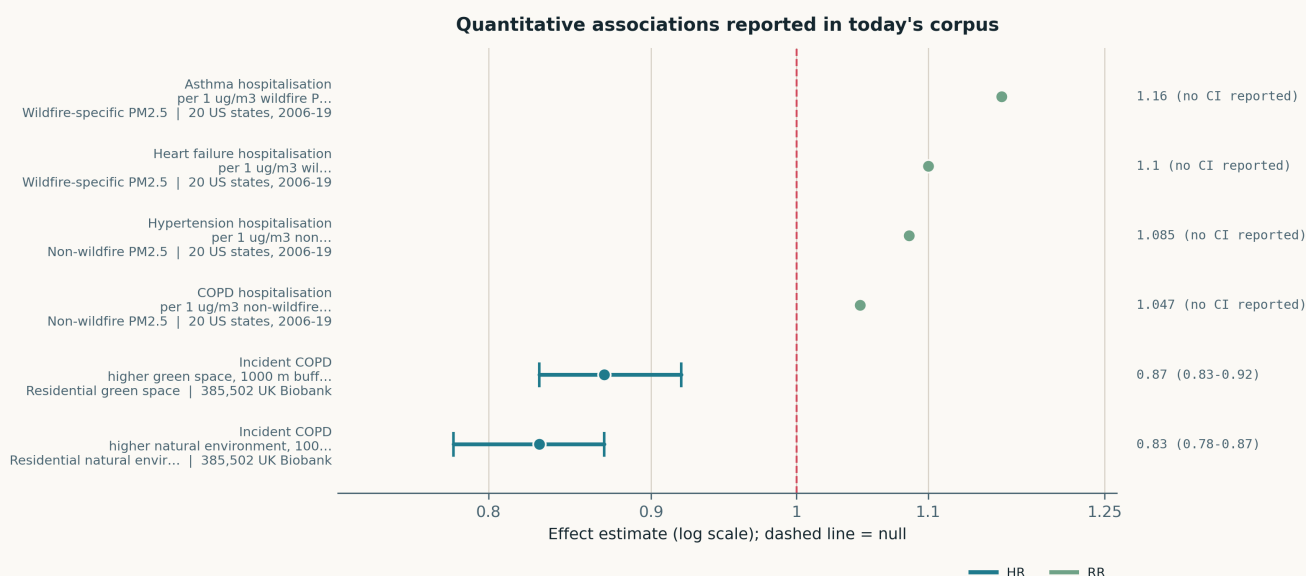


Fig. All six extractable effect estimates on a common logarithmic axis. The asymmetry is the finding: **the only two estimates with confidence intervals both sit below the null** and come from the same UK Biobank greenness analysis, while **all four estimates above the null are point values with no interval reported** — the wildfire and non-wildfire hospitalisation risks, whose abstract gives percentages without precision. Exposure contrasts are not mutually comparable: the hazard ratios are per greenness category at a 1000 m buffer, the risk ratios per 1 $\mu\text{g}/\text{m}^3$ of source-resolved $\text{PM}_{2.5}$. Read direction and precision, never relative magnitude.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro, cognition & mental health

1 record

Temporao et al. 2026 medRxiv (preprint)

Daily associations between outdoor air pollution and sleep quality

Budapest Sleep, Traits and Experiences Study — seven days per participant of daily activity and sleep-quality reports plus home EEG giving sleep efficiency, %REM, onset latency and total sleep time. Linear mixed models with participant random intercepts, adjusted for age, sex, BMI, education, weekday, season, temperature, physical activity, screen exposure and travel. **Day-1 ozone predicted lower day-2 sleep efficiency and poorer perceived quality**, surviving multiple-comparison correction, multi-pollutant models and sensitivity analyses for depression, insomnia and chronotype; **no robust association for PM_{2.5} or NO₂**. The null is confounded with exposure-model quality — city-aggregated station data give near-zero within-city contrast, and misclassification is worse for PM_{2.5} than for the more spatially homogeneous ozone. Sample size is not stated in the abstract.

doi: 10.64898/2026.09.15.26363130

Cardiovascular & metabolic

1 record

Zhang et al. 2026 Nat Commun

Cardiopulmonary hospitalisation risks from wildfire-specific and non-wildfire PM_{2.5} in 20 US states

Self-controlled design over 20 states, 2006–19, contrasting two-year average wildfire-specific against non-wildfire PM_{2.5}. Per 1 µg/m³, wildfire PM_{2.5} carried **10% (heart failure) to 16% (asthma)** higher hospitalisation risk; non-wildfire PM_{2.5} **4.7% (COPD) to 8.5% (hypertension)**. Associations were stronger among minorities, metropolitan residents, the less educated and more deprived communities. The toxicity contrast at equal mass is the contribution, and it undercuts the mass-only metric every regulatory network and low-cost sensor reports. Two limits: no confidence interval is given for any of the eight headline risks, so precision cannot be judged; and wildfire-specific PM_{2.5} is itself modelled, so the toxicity difference is conditional on a smoke-attribution step whose error is not propagated.

doi: 10.1038/s41467-026-76974-7

Respiratory & allergic

2 records

San Juan-Reyes et al. 2026 Environ Sci Pollut Res Int

Short-term ambient air pollution and respiratory hospitalisation in children under five

266 children under five admitted to one regional IMSS hospital in the Toluca Valley Metropolitan Zone during 2023; matched case-crossover with conditional logistic regression, primary window lag 1–7, adjusted for temperature and humidity. PM_{2.5} was associated with admission for viral pneumonia, bacterial pneumonia and asthma, CO with bronchiolitis, with heterogeneous municipality-level associations. **The abstract reports direction only — not one odds ratio, interval or unit appears** — so the effect size is unverifiable from the published summary. 266 events split across four diagnoses, five pollutants, several lag windows and three municipalities is thin, and exposure is entirely fixed-station with no personal or indoor component in a setting where indoor sources plausibly dominate a young child's dose.

doi: 10.1007/s11356-026-38237-2

Chen et al. 2026 Int J Chron Obstruct Pulmon Dis

Residential green and blue space, genetic susceptibility and incident COPD

385,502 UK Biobank participants, Cox models, greenness at 300 m and 1000 m buffers against a COPD polygenic risk score. At 1000 m: **green space HR 0.87 (95% CI 0.83–0.92), natural environment HR 0.83 (0.78–0.87)**; blue space weak and inconsistent. No multiplicative or additive interaction with genetic risk, and the natural-environment association held in every genetic stratum — so the protective signal is not running through a susceptible subgroup. The inferential problem is what is absent: **no air-pollution term is reported**. Residential greenness is inversely correlated with traffic PM_{2.5} and NO₂ nearly everywhere, so an unadjusted greenness–COPD association may simply restate a pollution–COPD association.

doi: 10.2147/copd.s627458

Mechanistic toxicology

2 records

Gao et al. 2026 Front Public Health

CCL22-associated immune signatures in air pollution-related atopic dermatitis

Shared pollutant–atopic dermatitis targets enriched in cytokine and chemokine signalling, pattern-recognition receptor activity, IL-17 and Toll-like receptor pathways. A framework over 15 algorithms and 175 combinations selected plsRglm + AdaBoost, with AUC 0.963 in training and 1.000, 0.909 and 0.966 across three validation cohorts, giving a ten-gene signature (CCL22, CCL5, CSF2, F2RL1, HRH4, ICAM1, IFNG, IL10, IL17A, IL18). CCL22 was upregulated in lesional samples, localised to dendritic cells and macrophages, and confirmed in DNFB-induced murine dermatitis. **The exposure link is the weak joint: no particulate matter was measured or administered anywhere**, so air pollution enters only as a database annotation. An AUC of exactly 1.000 indicates leakage or overfitting rather than performance, and DNFB is a contact hapten, not a particle.

doi: 10.3389/fpubh.2026.1906298

Vera-Colón et al. 2026 — metadata only Part Fibre Toxicol

Airborne fine particulate matter and human embryonic stem cell osteogenic differentiation

BMC deposited this DOI on 18 September with no abstract; Crossref, Europe PMC core, Europe PMC fullTextXML and the publisher landing page all returned nothing on per-DOI retry — **the first time this gap has appeared on a BMC rather than an Elsevier deposit**. Carried on title because skeletal development is an endpoint the PM literature has barely touched, and a human-embryonic-stem-cell osteogenic model would be a new mechanistic line rather than another airway or endothelial assay. No finding is asserted: neither particle source, dose, duration nor the direction of the reported disruption is verifiable from metadata. On the retry list.

doi: 10.1186/s12989-026-00701-9

Occupational & indoor exposure

2 records

Khan et al. 2026 Front Public Health*Cement industry emissions and respiratory health: global systematic review and meta-analysis, 2000–2025*

Eighteen observational studies, PROSPERO CRD420251268913. Random-effects pooling gave standardised mean differences of **−0.71 for FEV₁**, **−0.56 for FVC** and **−0.57 for FEV₁/FVC** in exposed groups; chronic cough was significantly more common while other symptoms were elevated but not significant. Occupational exposure produced greater FEV₁ impairment than residential — the expected dose ordering and the most reassuring internal check available. Reporting is the weakness: **no confidence interval, no I² and no prediction interval accompanies any pooled estimate**, so neither precision nor heterogeneity can be assessed, and SMDs pooled across studies using different spirometry reference equations with no common dust metric cannot be read as a dose–response.

doi: 10.3389/fpubh.2026.1887654

Assan et al. 2026 Health Sci Rep*Ambient particulate matter, urinary arsenic and resident health in two Ghanaian mining communities*

65 residents each from an artisanal and an incorporated mining community, all resident at least five years; PM by Modular Air Quality System, urinary arsenic by AAS, kidney function by albumin–creatinine ratio. More artisanal residents fell in the at-risk band for kidney disease ($p = 0.006$); the incorporated community showed poorer general health status. Descriptive only: $n = 130$, chi-square on dichotomised uACR with no confounder adjustment and no exposure–response, and **neither PM concentrations nor arsenic levels are numerically reported** despite being the two measured exposures. The 35 and 70 $\mu\text{g}/\text{m}^3$ figures quoted as WHO standards are the 2005-era interim targets, not the 2021 guideline values of 15 and 45 $\mu\text{g}/\text{m}^3$ for 24-hour PM_{2.5} and PM₁₀ — so the exceedance framing understates the gap.

doi: 10.1002/hsr2.73246

Sensing, forecasting & instrumentation

4 records

Bai et al. 2026 Atmos Meas Tech*Low-level jet evolution and vertical aerosol redistribution: a Doppler wind lidar study*

Two dust episodes over Hefei, East China. April 2021: the northwesterly jet core descended from 2.5 km to 0.3 km then re-ascended to 1.0 km while surface PM₁₀ peaked at **410 $\mu\text{g}/\text{m}^3$** , with shear above **0.04 s^{−1}** and increased spectral width at the lower jet interface and a downward shift of enhanced backscatter. March 2022: a stable jet core and persistent lower-tropospheric stability held aerosol aloft, the PM₁₀ peak arriving at 16:00 LT, **~8 h after inferred frontal onset**. Two regimes — rapid downward mixing during jet descent versus delayed surface impact from a decoupled layer. Two cases is mechanism illustration, not climatology, and the inertial-oscillation attribution rests on hodograph rotation alone. The clearest statement yet of why a surface-only network can read clean for hours.

doi: 10.5194/amt-19-5951-2026

Huang et al. 2026 Environ Sci Technol*Interpretable spatiotemporal coupling network for emission prediction in sludge gasification*

PISTC-Net couples a directed process topology with per-feature optimal time lags, reaching R^2 of **91.18% for SO₂**, **80.37% for NO_x** and **78.09% for PM** on industrial-scale data. Under simulated abrupt sensor failure it continued tracking emission trajectories via learned process inertia, and hierarchical attention shifted from upstream thermochemical zones for SO₂ to whole-process attention for PM, flagging disturbance propagation **5–8 minutes** before emission peaks. That PM is the worst-predicted species is the informative result: particle formation runs through nucleation, condensation and agglomeration paths a topology-plus-lag representation captures least well, whereas gas-phase sulfur is nearly a mass balance. Single plant, no external validation site, and interpretability is inferred from attention maps, which are a saliency artefact rather than a causal readout.

doi: 10.1021/acs.est.6c09135

Kalluri et al. 2026 — metadata only Atmos Environ*Long-term (2008–2024) black carbon over a semi-arid rain-shadow region in southern peninsular India*

The expected same-day Elsevier deposit gap: no abstract in Crossref, Europe PMC or PubMed on per-DOI retry. Carried on title because a 17-year single-site black-carbon record in a semi-arid rain-shadow setting is the kind of long baseline trend detection needs and South Asian networks mostly lack, and because absorption-coefficient records are what constrain the optical-to-mass conversion every filter- and scattering-based instrument depends on. No trend, magnitude or source attribution is asserted — the title names methods, not results. On the retry list.

doi: 10.1016/j.atmosenv.2026.122366

Krajnc Pogačnik et al. 2026 — metadata only Build Environ*Selective real-time monitoring of respiratory droplets for indoor air safety in educational environments*

Elsevier same-day deposit, abstract absent from Crossref, Europe PMC and PubMed on per-DOI retry. Retained because selective real-time discrimination of respiratory droplets against a background of ambient particles is the hard problem for any optical counter deployed indoors: a scattering sensor reports total number in a size bin and cannot by itself separate exhaled aerosol from resuspended dust, so any claimed selectivity mechanism is worth reading. Whether that selectivity is optical, chemical or inferential is not recoverable from the title, and no performance claim is asserted. On the retry list.

doi: 10.1016/j.buildenv.2026.115281

Exposure assessment & modelling

4 records

Kim et al. 2026 Environ Health

Residential indoor air monitoring for long-term exposure assessment in susceptible populations

130 older adults with idiopathic pulmonary fibrosis, Seoul Metropolitan Area, 2021–23. Indoor hourly $\text{PM}_{2.5}$ averaged **14.90 $\mu\text{g}/\text{m}^3$ (SD 36.12)** against **20.44 (61.63)** outdoors, higher in winter (20.17) than summer (11.19) and by day than night, with hourly patterns that contrast rather than track outdoors. Infiltration factors from median indoor/outdoor ratios of sulfur, nickel and iron were **51–70%**. Source contrasts were large: pets 24.09 vs 13.35, humidifiers 19.19 vs 13.16, smokers 16.76 vs 11.79, cooking 15.14 vs 10.96 $\mu\text{g}/\text{m}^3$. Every standard deviation exceeds its mean, so the distribution is episodic and a seasonal mean is a poor summary; five days per season is thin support for a long-term claim and no health estimate is reported yet. The transferable piece is the non-contact protocol — instruments redeployed by postal mail, which is how an unattended network is actually serviced.

doi: 10.1186/s12940-026-01334-3

Yu et al. 2026 Atmos Chem Phys

Continental pollutants modulate organic nitrogen and light absorption of marine organic aerosols

Shipboard sampling over the Yellow and Bohai Seas across three seasons. Aerosol organic nitrogen averaged **0.35 $\pm$ 0.25 $\mu\text{g N m}^{-3}$** , 4–60% of total nitrogen, highest in autumn. Anthropogenic secondary pollutants — aged biomass burning and secondary nitrate — supplied **36–76% of ON**, 46–83% of the water-soluble and 39–89% of the water-insoluble fraction; in spring dust accounted for 55% of ON and above 80% during a dust episode, while summer carried a marine biogenic component. Organic-aerosol light absorption rose with both ON concentration and ON/OC ratio, so absorption capability per unit carbon increased with nitrogen enrichment rather than tracking mass. Three cruises give no annual coverage and apportionment is receptor modelling on a small campaign. The sensing consequence is direct: an optical instrument calibrated on one absorption regime is misapplied when composition shifts.

doi: 10.5194/acp-26-13175-2026

Sunyer-Caldú 2026 Environ Sci Technol

Beyond blood and urine: decoding the human chemical exposome across matrices and time

A Perspective arguing that high-resolution mass spectrometry can profile thousands of chemical features per sample while study design has not kept pace — no investigation has combined external source monitoring, multi-matrix internal biomonitoring and toxicokinetic normalisation in one longitudinal framework. Three elements are proposed: multi-matrix sampling extending beyond blood and urine to accumulation archives, developmental matrices and external media; a Unified Exposure Scale placing heterogeneous matrices on a common toxicokinetically normalised dosimetric axis; and a two-track hazard-prioritisation pipeline. No data, no validation, and the Unified Exposure Scale is circular where it matters — it needs toxicokinetic parameters that do not exist for the molecular dark matter it is meant to rank. Logged for how air monitoring is being positioned inside the Global Human Exposome Project.

doi: 10.1021/acs.est.6c06509

Godri Pollitt 2026 — metadata only Environ Sci Technol

Beyond inhalation: air as an integrative matrix for exposome research

ACS deposited this DOI on 18 September without an abstract and no text was recoverable from Crossref, Europe PMC or PubMed on per-DOI retry. Carried because it is the **second air-as-exposome-matrix Perspective in this same issue**, alongside Sunyer-Caldú, which makes the pair an editorial signal rather than a coincidence: air sampling is being argued into the exposome programme as an integrative matrix rather than merely an inhalation route. No argument or claim is asserted beyond the title. On the retry list.

doi: 10.1021/acs.est.6c05073

Burden, policy & mitigation

1 record

Massimi et al. 2026 Environ Sci Technol

Urban green walls under traffic stress: elemental composition of PM deposition and plant response

A nine-species green wall at a high-traffic urban site, followed for a year. PCA on the insoluble fraction of Al, Cr, Cs, Fe, Mn, Mo, Rb, Sb, Sn, Ti and Zn identified *Photinia $\times$ fraseri* as the highest and most stable retainer of non-exhaust traffic particles and *Pteris vittata* as the lowest, with SEM-EDS showing **up to 100-fold higher retention** in the former. The methodological finding travels furthest: **only insoluble elements traced deposition reliably**, because soluble and whole-leaf fractions are dominated by endogenous pools of nutrients and of tracers such as Cs and Rb, so conventional whole-leaf digestion overestimates deposition. *Photinia* showed marked seasonal physiological and transcriptional modulation; *Pteris* did not. One wall at one site with no paired ambient PM mass measurement, so species are ranked against each other but no removal flux is quantified and no air-quality benefit can be inferred.

doi: 10.1021/acs.est.6c10086

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Temporao et al.	Daily associations between outdoor air pollution and sleep quality.	<i>medRxiv (preprint)</i>	Panel study with ambulatory EEG	Daytime outdoor PM _{2.5} , NO ₂ and O ₃ aggregated from city monitoring stations; same-day and previous-day	Hungary
Cardiovascular & metabolic					
Zhang et al.	Cardiopulmonary hospitalization risks from wildfire-specific and non-wildfire PM _{2.5} in 20 US states.	<i>Nat Commun</i>	Self-controlled design	Two-year average wildfire-specific vs non-wildfire PM _{2.5} , modelled and source-separated, per 1 µg/m ³	USA
Respiratory & allergic					
Chen et al.	Residential Green and Blue Space, the Natural Environment, Genetic Susceptibility, and Incident COPD.	<i>Int J Chron Obstruct Pulmon Dis</i>	Prospective cohort	Residential green space, blue space and natural environment at 300 m and 1000 m buffers; no PM term reported	UK
San Juan-Reyes et al.	Short-term effects of ambient air pollution exposure on respiratory hospitalizations among children in the State of Mexico.	<i>Environ Sci Pollut Res Int</i>	Case-crossover	Daily PM _{2.5} , PM ₁₀ , CO, NO ₂ and O ₃ from fixed monitoring stations; primary window lag 1-7 cumulative	Mexico
Mechanistic toxicology					
Gao et al.	Integrative transcriptomic analysis identifies CCL22-associated immune signatures in air pollution-related atopic dermatitis.	<i>Front Public Health</i>	Integrative transcriptomics + murine validation	No measured PM; pollutant-associated gene sets taken from a toxicogenomic database and intersected with atopic-dermatitis transcriptomes	Not stated
Vera-Colon et al.	Exposure to airborne fine particulate matter disrupts human embryonic stem cell osteogenic differentiation.	<i>Part Fibre Toxicol</i>	Metadata only (no abstract)	PM composition, dose and collection route not recoverable from metadata	Not stated
Occupational & indoor					
Assan et al.	Evaluation of Ambient Air Particulate Matter, Urine Arsenic Levels and the Effect on Residents Health in an Artisanal Mining Community and an Incorporated Mining Community in Ghana.	<i>Health Sci Rep</i>	Cross-sectional (comparative community)	Ambient PM _{2.5} and PM ₁₀ by Modular Air Quality System; urinary arsenic by atomic absorption spectrophotometry	Ghana
Khan et al.	Exposure to cement industry emissions and respiratory health outcomes: a global systematic review and meta-analysis from 2000 to 2025.	<i>Front Public Health</i>	Systematic review + random-effects meta-analysis	Cement dust and cement-plant emissions, occupational and residential; no harmonised gravimetric metric across studies	Global / multi-region
Sensing, forecasting & instrumentation					
Bai et al.	Role of low-level jet evolution in vertical aerosol redistribution: a Doppler wind lidar study over East China.	<i>Atmos Meas Tech</i>	Remote sensing / lidar retrieval	Coherent Doppler wind lidar backscatter, spectral width and shear profiles against surface PM ₁₀	China

First author	Focus	Journal	Design	Metric	Region
Huang et al.	Interpretable Spatiotemporal Coupling Network for Robust Emission Prediction and Implicit Mechanism Mining in Sludge Gasification.	<i>Environ Sci Technol</i>	Deep-learning forecasting	Continuous industrial stack monitoring of PM, SO ₂ and NO _x from a sludge-gasification plant, with feature-level optimal time lags	Not stated
Kalluri et al.	Long-term (2008-2024) analysis of black carbon aerosols over a semi-arid rain-shadow region in southern peninsular India: trends, wavelet analysis, machine learning approaches and source regions.	<i>Atmos Environ</i>	Metadata only (no abstract)	Black carbon, presumably aethalometer-derived; instrument, wavelength set and trend magnitudes not recoverable	India
Krajnc Pogacnik et al.	Selective real-time monitoring of respiratory droplets for indoor air safety in educational environments.	<i>Build Environ</i>	Metadata only (no abstract)	Respiratory-droplet detection; sensing principle, size range and selectivity mechanism not recoverable	Not stated
Exposure assessment & modelling					
Godri Pollitt	Beyond Inhalation: Air as an Integrative Matrix for Exposome Research.	<i>Environ Sci Technol</i>	Metadata only (no abstract)	No measured PM; air framed as an exposome sampling matrix	Global / multi-region
Kim et al.	Residential indoor air monitoring to assess long-term exposure to air pollution in susceptible populations for epidemiology: overview and early findings.	<i>Environ Health</i>	Long-term indoor monitoring campaign	Indoor and matched outdoor PM _{2.5} plus 36 chemical components, RTI MicroPEM, 5-day runs in each of four seasons, hourly resolution	South Korea
Sunyer-Caldu	Beyond Blood and Urine: Decoding the Human Chemical Exposome across Matrices and Time.	<i>Environ Sci Technol</i>	Perspective / framework proposal	No measured PM; air positioned as one external matrix within a multi-matrix exposome design	Global / multi-region
Yu et al.	Continental pollutants modulate organic nitrogen and light absorption of marine organic aerosols over East Asian marginal seas.	<i>Atmos Chem Phys</i>	Field campaign / shipborne	Shipboard aerosol organic nitrogen, water-soluble and water-insoluble fractions, ON/OC ratio and light absorption over three seasons	China
Burden, policy & mitigation					
Massimi et al.	Urban Green Walls under Traffic Stress: Linking the Elemental Composition of Particulate Matter Deposition to Plant Physiological and Gene Expression Responses.	<i>Environ Sci Technol</i>	Biomonitoring + source apportionment	Multi-elemental fractionation of PM deposited on green-wall foliage - insoluble, soluble and whole-leaf fractions - with SEM-EDS, over one year	Italy

Provenance and method

Entry window **18 September 2026**, single day, continuous with the previous issue. Legs run: **PubMed** E-utilities on the health and instrumentation axes as separate queries (3 unique records; the PubMed connector returned the same 4 PMIDs, of which 3 were in the local harvest); **Europe PMC** **CREATION_DATE** (44); **Crossref by ISSN** (24). **OpenAlex** returned HTTP 429 and **arXiv** HTTP 406, both known-broken legs. 71 raw records deduplicated to 69 on PMID → DOI → normalised title, 9 dropped as already carried in a previous issue, leaving 60 fresh candidates of which **17 were admitted**. Of the 43 excluded, **4 were Europe PMC retro-index artefacts** — PMC-sourced records returned inside the window whose first publication dates are 2025-07-23, 2026-01-01, 2026-07-17 and 2026-08-28. This extends the AGRICOLA retro-index defect logged on 16 September to the PMC source; a first-publication-date gate at the issue month is now applied to abstract-less PMC records. **Consensus** returned 10 sensor-calibration papers and admitted **none**: 5 matched archive titles, and all 5 survivors failed a Crossref **created** check (2026-01-17 to 2026-06-15). **Scholar Gateway** was queried once for abstract recovery and returned only off-topic Wiley content, consistent with prior runs. Four records are carried on **metadata alone** and assert no finding. All 17 DOIs were resolved against Crossref before build. Subtopic, design, particle-metric and geography labels are assigned from title, abstract, keywords and MeSH; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Effect estimates are transcribed verbatim and use heterogeneous exposure contrasts — see the forest-plot caption. Abstracts remain the copyright of their respective publishers.